

# Build the Recommender

Project: Transparent Book Recommender

## 1. Build the Recommender

### Teach the learning cycle

Primary target: Implement a deterministic recommender

#### Session · Sequence and minutes · Resources

**Session:** Session 1 · Understand · 45 min

**Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Case exposure check: this source example appears in ai-2/24, ai-2/33. Reuse a familiar case for recall or

modelling; verify that the case used to assess transfer has not already received feedback.

## Retrieval answer guidance

### Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

Look for: Define a small application purpose

Reference example, not the only acceptable wording: Suggest a suitable eligible activity for a teacher-supervised session; the teacher reviews and decides.

### Agents and Tools

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of state machines for bounded agents

Reference example, not the only acceptable wording: Teacher explanation: A bounded agent uses explicit states and finite action limits. A state machine makes invalid transitions visible. States are ready, looking\_up, review, done. ready may start lookup; lookup may finish done or request review; review may finish done after authorised evidence. done is terminal. Maximum two lookup attempts.

### Predictive vs Generative

Without opening the old lesson, explain one key idea from Predictive vs Generative. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish discriminative prediction from generation by inspecting the requested output

Reference example, not the only acceptable wording: Teacher explanation: Message labels are discriminative prediction; continuing a sequence is generation; record lookup is retrieval rather than learned generation by itself. One application can route a request, retrieve evidence and generate a response.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

### What the learner must demonstrate

Catalogue: A mystery/90/illustrated, B mystery/140/not illustrated, C science/80/illustrated. Request mystery, maximum 100 pages. Rank with 2 genre points and 1 illustration point after filtering. Show why other books are excluded.

### Concept to explain

A transparent baseline makes later improvements understandable.

## Term · Meaning

**Term:** baseline

**Meaning:** A simple reference method used to judge whether a more complex method adds value.

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**Term:** candidate

**Meaning:** An item or proposed answer being considered before a final selection.

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**Term:** weighted total

**Meaning:** A sum formed after multiplying each contribution by its assigned weight.

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**Term:** explanation

**Meaning:** An account connecting a result to the mechanism, calculation or evidence that supports it.

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## Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

## Preparation for the connected explanation

Teach Week 24 as a two-session investigation using fictional data. Build a fictional activity recommender using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name. This makes the outcome reproducible rather than dependent on input order. Assume unique names and validated numeric fields. Assess the three stated outcomes through the independent task, not by copying the worked result.

## Readiness answer

No. Filter by the hard constraint before ranking.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

## Apply the connected idea: Building the data pipeline

Teach Week 32 as a two-session investigation using fictional data. Build the validation stage of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5. The batch stage must separately check name uniqueness. Only accepted records may reach the Week 24 selection function. Assess the three stated outcomes through the independent task, not by copying the worked result.

### Readiness answer

Validate, filter eligibility, calculate scores, resolve ties, then report with limits.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional input records from this lesson; no real student identifiers

### End goal for the pictorial case

By the end, I can implement deterministic filtering, scoring and ties for a complete catalogue case.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

## 2. Explain the mechanism

### Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. Weights: genre=3, difficulty=2, length=1.

2. Compute every candidate's terms and total.

3. Filter hard constraints, sort remaining totals high to low.

4. Output top choice, score breakdown, one limitation, and a tie rule.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### Concrete teaching case

A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Compute the winner and evidence to display.

## Exact explanation to model

A wins at 3; B=2; C=1. Show flags, formula, scores and explanation; declare tie policy.

## Explain the added worked example

Build a fictional activity recommender using last week's score. Require duration at most a supplied limit. On a tie choose the alphabetically first name. This makes the outcome reproducible rather than dependent on input order. Assume unique names and validated numeric fields.

The function prints nothing; the caller prints A. best starts as None to mean no eligible option found. Python short-circuits or: when best is None, later comparisons in that condition need not run. After selecting an item, best\_score is numeric. An empty list or no eligible item returns None.

## Key terms

### Term · Meaning

**Term:** Recommender

**Meaning:** A procedure that suggests options using a stated policy

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**Term:** Tie

**Meaning:** Two options have the same comparison score

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**Term:** Deterministic

**Meaning:** The same inputs and rules produce the same result

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**Term:** Explanation

**Meaning:** Evidence describing how a result was obtained

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## Apply the connected idea: Building the data pipeline

Build the validation stage of the planned chooser first. A valid record has nonempty text name, integer minutes 0 through 60 and integer fun and learning ratings 0 through 5. The batch stage must separately check name uniqueness. Only accepted records may reach the Week 24 selection function.

The worked calls print True then False. Short-circuit checks keep absent or wrongly typed fields away from later operations. This Boolean helper does not explain the reason; add reason codes in the reporting stage

or use a validation result record. Do not describe a range failure as a no-match outcome.

The first passes. Replacing minutes with 61 fails the range. Exit: No. The batch uniqueness contract fails. Block the ambiguous batch and report duplicate names for correction.

## Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

**Predict the result if the list order is reversed, and if the limit is 8.**

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### Thinking move · Teacher prompt

**Thinking move:** Plan

**Teacher prompt:** What is given? What is the question? Which procedure fits these facts?

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**Thinking move:** Do

**Teacher prompt:** Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

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**Thinking move:** Check

**Teacher prompt:** Does the output answer the question? Which assumption or limitation prevents a larger claim?

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Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

## A pictorial worked case: Build the Recommender

A fictional book chooser filters a request before comparing preference scores.

The facts we will use

Request: animal subject, at most 90 pages. Score 2 for illustrated and 1 for comic. Tie chooses smaller ID.  
A animal90 illustrated prose; B animal70 plain comic; C space60 illustrated comic; D animal80 illustrated comic.

AI ACADEMY / AI-2/33

Filtering before scoring

|   | Pages | Preference score | Status        |
|---|-------|------------------|---------------|
| A | 90    | 2                | eligible      |
| B | 70    | 1                | eligible      |
| C | 60    | 3                | subject fails |
| D | 80    | 3                | eligible      |

Read a cell using both its row and column labels.

## Build the Recommender

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Filter

A,B,D meet subject and page requirements; C fails subject.



#### 2. Compute scores

A=2; B=1; D=3.



#### 3. Choose best

Return D with both preference matches recorded.



#### 4. Deterministic output

Reordering the input rows must not change D's unique winning score.

### CHANGE ONE THING

Change D's pages to 91, leaving every other field unchanged.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Filter

**What happens and why:** A,B,D meet subject and page requirements; C fails subject.

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**Step:** 2. Compute scores

**What happens and why:** A=2; B=1; D=3.

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**Step: 3. Choose best**

**What happens and why:** Return D with both preference matches recorded.

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**Step: 4. Deterministic output**

**What happens and why:** Reordering the input rows must not change D's unique winning score.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Eligibility and ranking answer different questions. C has a high preference score but is not an animal book, so it must stay excluded. The implementation needs a no-match result and explicit tie policy even when this example has a unique winner. Save rejected reasons for an explanation, not just the final ID.

### What this example does not establish

Toy preference weights do not imply shorter or illustrated books are universally better for all learners.

### Change one thing in the pictured case

Change: Change D's pages to 91, leaving every other field unchanged.

Prediction: D becomes ineligible, so A wins with score 2.

Reason: The strict excess over the 90-page maximum changes filtering before ranking.

### Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

## 3. Demonstrate and guide practice

### Demonstrate and guide practice

#### Board sequence

Write the given facts: A has flags (1,1), B=(1,0), C=(0,1); score=2g+l.

Write the question to explain: Compute the winner and evidence to display.

Reveal after predictions: A wins at 3; B=2; C=1. Show flags, formula, scores and explanation; declare tie policy.

Keep for an independent attempt: A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

## Classroom dialogue

**Ask:** Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

**Say:** Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

## Model these guided questions

Reversing still returns A. Limit 8 leaves no eligible item and returns None. Exit: X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

## Case-specific teaching sequence

### Stage · Teacher action

**Stage:** Session 1 start to 7 minutes

**Teacher action:** Retrieve the prerequisite and explain the purpose in the opening section.

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**Stage:** 7 to 20 minutes

**Teacher action:** Read the worked example and diagram; ask learners to trace each step aloud.

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**Stage:** 20 to 33 minutes

**Teacher action:** Complete W1 and W2, then compare predictions with observed results.

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**Stage:** 33 to 45 minutes

**Teacher action:** Collect the guided prediction before running; discuss a mismatch and its cause.

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**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the main rule without showing the previous answer.

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**Stage:** 8 to 20 minutes

**Teacher action:** Use W3 to diagnose and repair the stated failure; rerun an earlier case.

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**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently, including code or plan, predictions and explanations, before hints.

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**Stage:** 35 to 45 minutes

**Teacher action:** Discuss W5 and the exit; use the fresh retry when support was needed.

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## Teach the connected idea

### Stage · Teacher action

**Stage:** Session 1 start to 7 minutes

**Teacher action:** Retrieve the prerequisite and explain the purpose in the opening section.

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**Stage:** 7 to 20 minutes

**Teacher action:** Read the worked example and diagram; ask learners to trace each step aloud.

---

**Stage:** 20 to 33 minutes

**Teacher action:** Complete W1 and W2, then compare predictions with observed results.

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**Stage:** 33 to 45 minutes

**Teacher action:** Collect the guided prediction before running; discuss a mismatch and its cause.

---

**Stage:** Session 2 start to 8 minutes

**Teacher action:** Retrieve the main rule without showing the previous answer.

---

**Stage:** 8 to 20 minutes

**Teacher action:** Use W3 to diagnose and repair the stated failure; rerun an earlier case.

---

**Stage:** 20 to 35 minutes

**Teacher action:** Collect W4 independently, including code or plan, predictions and explanations, before hints.

---

**Stage:** 35 to 45 minutes

**Teacher action:** Discuss W5 and the exit; use the fresh retry when support was needed.

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## Reduce help deliberately

Completion task: State `recommend([], 10)` and the worked items with limit 8.

Teacher check: Both return `None`; the caller should display no eligible option.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

## Individual reasoning check

Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

## Connected teaching cycle: Building the data pipeline

Target: Separate validation selection and reporting

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: What does `valid_item` return if learning is absent?

Teacher check: `False`: `get` returns `None` and the exact integer check fails.

Individual check: For Building the data pipeline, use its worked case to evaluate this proposed explanation:

“A successful run proves the conclusion”. Explain the deciding step.

Expected correction: Each stage should have a clear input and output contract. A program can be correct at one stage but wrong at their connection, such as passing rejected rows into selection. Keep acceptance and rejection counts in the report.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

### Pictorial practice question

Explain why B does not win after D is excluded even though it has fewer pages.

### Reasoned teacher answer

Page count is an eligibility limit, not a ranking preference here. A scores 2 and B scores 1.

Changed-case reasoning: D becomes ineligible, so A wins with score 2. The strict excess over the 90-page maximum changes filtering before ranking.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

## 4. Run the weekly evidence lab

### Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Build the editable score table, hand-check one row, run two profiles, and compare system ranking with independent recalculation.

What did the hardest test teach you that the normal test did not?

### Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

#### Record · What to capture

**Record:** Input or evidence

**What to capture:** Exact data, instruction or claim

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**Record:** Expected result

**What to capture:** Answer or safe fallback declared before the run

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**Record:** Observed result

**What to capture:** Actual result, including errors

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**Record:** Explanation

**What to capture:** Rule, calculation or evidence explaining the outcome

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**Record:** Next test

**What to capture:** One justified change and a new prediction

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## Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

## Address these additional misconceptions

A successful run proves the conclusion: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

A convenient shortcut preserves meaning: Updating best on score  $\geq$  best\_score makes the last tied item win, so reordering can change results. Use the explicit name comparison for ties. Test both orders and a no-match case after the repair.

## Check these misconceptions

A successful run proves the conclusion: Each stage should have a clear input and output contract. A program can be correct at one stage but wrong at their connection, such as passing rejected rows into selection. Keep acceptance and rejection counts in the report.

Matching the example is enough: Require the new W4 case, explicit predictions and an explanation. Fresh

retry: X (minutes 9, fun 2, learning 3), Y (10,3,2), Z (11,6,1), limit 10. Accept X/Y, reject Z rating; select Y score 8 versus X 7.

A convenient shortcut preserves meaning: Calling the selector on raw data can crash or produce a result from invalid ratings. Route data through validation first. Duplicate names require a batch check even when each row individually passes; block the ambiguous batch for review instead of arbitrarily keeping one.

## Match the support to the first error

### What the response shows · Next teaching action

**What the response shows:** Missing or misread facts

**Next teaching action:** Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

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**What the response shows:** Wrong procedure

**Next teaching action:** Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

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**What the response shows:** Arithmetic or implementation error

**Next teaching action:** Trace intermediate values or lines; preserve the first mismatch and retry that step.

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**What the response shows:** Correct result without a reason

**Next teaching action:** Ask for a trace and a changed case before marking independent understanding.

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**What the response shows:** An unsupported wider claim

**Next teaching action:** Ask which supplied evidence supports the claim and which further observation would be needed.

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Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

## 5. Assess and explain the answers

### Assess and explain the answers

## Worked case answer

A wins at 3; B=2; C=1. Show flags, formula, scores and explanation; declare tie policy.

## Independent question

A reader requests space and at most 60 pages. Book A is space, 90 pages; B is animals, 40 pages. What should the prototype return if both conditions are required?

## Expected independent answer

Neither matches both conditions. Return no exact match and ask whether to relax a preference; do not present A or B as a full match.

## What counts as evidence

**Criterion · 0 points · 1 point · 2 points**

**Criterion:** Result

**0 points:** Incorrect or absent

**1 point:** Partly correct

**2 points:** Correct with necessary conditions

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**Criterion:** Reasoning

**0 points:** No supporting trace

**1 point:** Some steps

**2 points:** Clear mechanism, calculation or source support

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**Criterion:** Limit

**0 points:** Overclaims certainty

**1 point:** Vague limitation

**2 points:** Relevant uncertainty or missing fact

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**Criterion:** Transfer

**0 points:** Copies without adapting

**1 point:** Adapts with prompts

**2 points:** Solves a changed case independently

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Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

## Feedback

Ask the learner to find the evidence that challenges this claim: “The system must recommend something even when constraints fail.” Keep the first attempt and an uncoached retry separate.

## Original transfer answer

Only A meets genre and length constraints; its score is 3. B exceeds the page limit and C fails genre. Do not claim A beat eligible competitors when none existed. Include no-match and missing-field handling in the prototype.

## Explanations for the additional workbook examples

### W1 Trace selection

Why does A win the worked example even though B is processed first?

Both score 8 and qualify; A is alphabetically first under the tie rule.

### W2 Handle no match

State `recommend([], 10)` and the worked items with limit 8.

Both return `None`; the caller should display no eligible option.

For the empty list, the loop has no iterations, so `best` stays `None`. With the worked durations 10 and 9 under a limit of 8, neither item passes the eligibility check, so `best` again stays `None`. The caller must turn this result into a clear no-eligible-option message; it must not treat `None` as an activity name.

### W3 Repair order dependence

A tied selection changes after reversing input order. Identify the likely missing policy and test.

An explicit deterministic tie rule is missing or incorrect. Apply alphabetical name order and rerun both orders.

### W4 Implement independently

Use the function with X (fun 4, learning 1, minutes 12), Y (3,3,10), Z (5,5,13), limit 12. Predict the winner, reverse the list and retest. Also test limit 9.

X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns `None`.

## W5 Explain the result

Write a one-sentence explanation of X winning without calling it objectively best.

X meets the 12-minute limit and ties for highest eligible score 9; the alphabetical tie rule selects X.

## Building the data pipeline — W1 Trace a validation

Explain why a row with fun "4" fails while fun 4 can pass.

The first is text; the schema requires an exact integer rating from 0 to 5.

## Building the data pipeline — W2 Check missing fields

What does `valid_item` return if learning is absent?

False: `get` returns None and the exact integer check fails.

When learning is absent, `row.get("learning")` returns None. The condition `type(value) is not int` is true, so the helper returns False before doing the range comparison. This is a schema rejection; it is different from a valid item that later exceeds the selection time limit.

## Building the data pipeline — W3 Repair the connection

A program validates rows but still sends the original raw list to recommend. What should change?

Pass only the accepted, uniqueness-checked list. Preserve rejection information separately for reporting.

## Building the data pipeline — W4 Build independently

Combine validation and the Week 24 selector for A (minutes 10, fun 3, learning 2), B (12,4,1), and C (8,"5",5), limit 10. State accepted rows, rejected row, selection and explanation.

A and B validate; C fails fun type. B exceeds the limit. A wins with score 8. Report 2 accepted of 3, one rejection and one eligible item. Verify unique names.

## Building the data pipeline — W5 Handle duplicates

Two valid rows have the same name but different ratings. Should individual validation alone permit selection?

No. The batch uniqueness contract fails. Block the ambiguous batch and report duplicate names for correction.

## Assess independent transfer

Selected case: Request: mystery, at most 100 pages. Catalogue: D mystery110 illustrated; E mystery80 plain; F science80 illustrated; G mystery95 illustrated; H mystery90 illustrated. Filter first; score 3 for mystery plus 2 for illustrations; break ties by fewer pages.

## Selected case answer

D fails length and F fails genre. E scores 3, G and H score 5. H wins the tie with 90 versus 95 pages. Filtering is separate from scoring; this written preference rule is not proof of a trained model.

### Dimension · Evidence required

**Dimension:** Explanation

**Evidence required:** Names the applicable idea or procedure and ties it to the supplied facts.

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**Dimension:** Trace and result

**Evidence required:** Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

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**Dimension:** Limits

**Evidence required:** States the assumptions, missing evidence or conditions under which the result would change.

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Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

## Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

# 6. Connect and extend learning

## Connect and extend learning

### Transparent Book Recommender

This is the complete baseline: inputs, scoring, filter, ranking, explanation, and abstain rule.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record. Retrieve this record next week rather than restarting the project.

## Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

## Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

## Bridge to the next week

Next: Week 34: Test Diverse Users. Retrieve today's evidence and identify what carries forward and what is new.

## Support and fresh reassessment

Trace best and best\_score row by row. Extension: return a record with name, score and reason while retaining the same selection policy.

Independent W4 evidence: X and Y score 9; Z is excluded. X wins alphabetically in either order. Limit 9 returns None. Record whether the learner needed no help, a hint or explanation. Fresh retry: B (2,4,8), C (3,2,8), A (5,5,9), limit 8. B/C tie at 8; B wins. Limit 7 returns None.

Next week: File input and output. Reuse this week's explicit contracts and evidence.

## Readiness and reassessment for Building the data pipeline

Connect paper stage cards with input/output labels. Extension: return specific reasons for missing, type and range errors without exposing personal data.

Independent W4 evidence: A and B validate; C fails fun type. B exceeds the limit. A wins with score 8. Report 2 accepted of 3, one rejection and one eligible item. Verify unique names. Record whether the learner needed no help, a hint or explanation. Fresh retry: X (minutes 9, fun 2, learning 3), Y (10,3,2), Z (11,6,1), limit 10. Accept X/Y, reject Z rating; select Y score 8 versus X 7.

## Plan the delayed return

Suggested return points in this level: ai-2/34, ai-2/36. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

### Record · Teacher evidence

**Record:** Learner code and date

**Teacher evidence:** \_\_\_\_\_

**Record:** Concept and case used; prior exposure

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Explanation / trace / limits

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Support and first repair

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Changed case and delayed outcome

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

**Record:** Next teaching action

**Teacher evidence:** \_\_\_\_\_

\_\_\_\_\_

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

### Project evidence from this case

Save the recommender code, contract and this full trace; test the same records in different input orders.

## 7. Prepare, teach and assess

Teaching target: Implement a deterministic recommender

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

### Session · Plan · Resources

**Session:** Session 1 · Understand · 45 min

**Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

**Resources:** Sections 1–6; record the brief recall in the workbook.

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**Session:** Session 2 · Practise · 45 min

**Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

**Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

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**Session:** Session 3 · Investigate and transfer · 45 min

**Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

**Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

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Before class, solve the supported practice: `State recommend([], 10)` and the worked items with limit 8.

Reasoned practice answer: Both return None; the caller should display no eligible option.

Check every learner before discussion: Use the worked case for Implementing a recommender. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The recommendation follows a written scoring policy. It does not learn preferences, prove learning benefit or authorize an action. Display the chosen rule and let a person review whether an option is suitable.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change D’s pages to 91, leaving every other field unchanged. Require a prediction and a reason before comparing. Expected result: D becomes ineligible, so A wins with score 2. Reason: The strict excess over the 90-page maximum changes filtering before ranking.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

**Criterion · Not yet · With support · Independent**

**Criterion:** Explanation

**Not yet:** Names an answer without a deciding fact

**With support:** Uses a hint to link fact and decision

**Independent:** Explains the deciding fact in own words

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**Criterion:** Trace or test

**Not yet:** Cannot connect input and result

**With support:** Completes steps with prompts

**Independent:** Traces or tests a changed case accurately

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**Criterion:** Limits and revision

**Not yet:** Claims more than evidence supports

**With support:** Names a limit after a prompt

**Independent:** States a defensible limit and useful next test

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Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

## 8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.